

Juicetification: Aggregate Anxiety

Build a 12-month production plan for a juice bottler - student quick-start

YOUR MISSION

You are the operations analyst at Juicetification Inc. Demand peaks in spring and summer. Build a CHASE plan and a LEVEL plan by hand, compare them, then recommend one. Everyone gets different numbers - the method is the same.

CHASE

Match capacity to demand each month. Hire/fire (or flex hours) to follow the curve. Low inventory, but lots of workforce change.

LEVEL

Hold output steady all year; hire once in January, then coast. Stable workforce, but inventory builds and shortages can appear at the peak.

THE ROADMAP (11 stages, top to bottom in the sidebar)

0

Brief

1

Understand

2

Pick data

3

Columns

4

Formulas

5

Practice

6

Capacity

7

Chase

8

Level

9

Compare

10

Design

11

Report

USING THE WORKSHEET

- **Type a number or a formula**
e.g. `=D1/RATE` or `=C1+D1-B1`
- **Use column letters + row #**
row 1 = January, B = Forecast Demand
- **Named values work in formulas**
LABOR, HIRE, HOLD, RATE, BEGIN, SAFETY ...
- **Live review colors your cells**
red = wrong, amber = blank
- **Watch which 3 months show**
the review scrolls with the grid
- **Enter the total annual cost**
sum every month's Total Monthly Cost

KEY FORMULAS

- > $\text{Capacity/wkr} = 5/\text{hr} \times 8 \text{ hr} \times 25 \text{ d} = 1,000/\text{mo}$
- > Reg Production = demand adj. for inventory
- > $\text{Workers} = \text{ROUNDUP}(\text{Production} / 1,000)$
- > $\text{Hours/Day} = \text{Workers} \times 8$ (partial: hire hours)
- > $\text{Hires} = \text{MAX}(0, \text{this month} - \text{last month})$
- > $\text{Ending Inv} = \text{MAX}(0, \text{Begin} + \text{Prod} - \text{Demand})$
- > $\text{Backorders} = \text{MAX}(0, \text{Demand} - \text{Begin} - \text{Prod})$
- > $\text{Level Rate} = (\text{Total} + \text{end target} - \text{Begin})/12$
- > $\text{Total} = \text{Labor} + \text{Hire} + \text{Layoff} + \text{Hold} + \text{Backorder}$

COST SHEET

Labor \$3,200/wkr/mo Hire \$600 Layoff \$900 Holding \$0.25/bottle Backorder \$1.50 Overtime \$4.50 Begin 2,400

Stuck? Open "I'm stuck" to see the full worked solution - then it hands you a FRESH scenario so you rebuild it and prove you learned it.

Run with: `streamlit run app.py` - `v1` for publication